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Evaluation of high-strength concrete for lunar applications: Mechanical behaviour under static and dynamic loads at cryogenic temperatures

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ABSTRACT

Substantial advancements in human sciences have greatly accelerated the exploration of celestial bodies. The construction of lunar bases now faces unique and severe environmental challenges, requiring materials that ensure structural integrity and stability. This study examines the static and dynamic mechanical properties of an innovative cement-free lunar high-strength concrete (LHSC). The multi-physics coupling effects of the extreme temperatures (20 °C, −70 °C, and −170 °C) and varying strain rates (approximately 30–200 s^{-1}) were considered to evaluate the dynamic responses of LHSC using a 100 mm-diameter impact apparatus. The findings indicated that the compressive strength of LHSC increased from 129.5 to 153.6 MPa as the temperature decreased, with dynamic strength rising up to 70.8% above static levels at −170 °C. The energy absorption capacity generally improved with strain rate but signally deteriorated under cryogenic conditions. Dynamic split-tensile strength showed a consistent linear increase, with strength-rate sensitivity peaking at −70 °C whereas slightly diminishing at −170 °C. The ice-induced stiffening contributed to heightened material brittleness, leading to abrupt fracture initiation. The dynamic increase factor (DIF) for both compression and split-tension was higher at lower temperatures, highlighting increased rate sensitivity under cryogenic conditions, with split-tensile DIF values exceeding those of compression across all temperatures. Computed tomography (CT) scanning revealed shrinking pores and expanding ice crystals inside LHSC, which strengthened its matrix but also intensified microcrack propagation, reducing the material deformability. These insights are instrumental in developing the construction material specifically designed for the harsh lunar environments, thereby supporting strategies for sustainable extraterrestrial habitation.

1. Introduction

The Moon represents the principal frontier for human extraterrestrial exploration and is poised to evolve into a strategic base for future space endeavours. As the initiative continues to maintain human habitats on the Moon, the demand for robust construction materials intensifies. Concrete, extensively employed as a building material on Earth due to its proven durability and structural integrity, emerges as an ideal candidate for lunar construction projects [1,2].

Among the diverse classifications of concrete, alkali-activated concrete (AAC), also known as geopolymer concrete, presents a significant advancement over traditional cement-based systems. A key innovation lies in its ability to utilise the mineral components available in lunar regolith, thereby eliminating the need for conventional cement and aligning with the principles of in-situ resource utilisation (ISRU) [3–5]. Moreover, AAC boasts superior properties such as enhanced heat resistance, chemical durability, vacuum stability, and a notably reduced demand for water [6–9]. Unlike OPC systems, which irreversibly consume mixing water through hydration, AAC predominantly uses water as a reaction medium rather than forming chemically bound hydrates. Recent studies [10,11] have proposed the adoption of vacuum-curing and condensation systems to recover unreacted water from AAC mixtures, with reported recovery rates reaching 98% under simulated lunar conditions. These features make AAC highly advantageous for fabrication and application in lunar environments, where water is among the most critically limited resources.

On the Moon, environmental conditions like the significantly lower temperatures and prominent temperature fluctuations inevitably

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influence the various properties of AAC. The lunar surface undergoes drastic temperature variations, ranging from −173 °C during the lunar night to 127 °C in the daylight. Additionally, some areas at the poles are exposed to temperatures around −233 °C [12]. Xiong et al. [13] prepared geopolymer-based lunar concrete utilising various lunar regolith simulants and assessed its performance under extreme temperatures. The findings revealed that high temperatures on the lunar surface minimally impacted the concrete compressive strength. However, at −30 °C, the strength could reach up to 80 MPa, attributed to the combined strength of concrete matrix and ice. Furthermore, after 40 thermal shock cycles ranging from −196 °C to 25 °C, the strength exceeded 60 MPa, with microstructural analysis confirming that the matrix remained densely structured. Pilehvar et al. [14] studied the impacts of temperature fluctuations and vacuum exposure on 3D-printed lunar geopolymers using DNA-1 regolith simulant. They found that after experiencing one lunar day-night cycle, with temperatures between −80 °C and 114 °C, the compressive strengths of all samples increased as compared to those pre-cured for 3 h at 80 °C. The strength gain after lunar cycle was significantly greater under ambient conditions. Moreover, the geopolymers cured in vacuum conditions exhibited minimal mass loss, indicating slight susceptibility to the extreme lunar environment. However, Zhang et al. [15] and Wang et al. [11] observed detrimental effects of the extreme temperature variations on lunar geopolymer properties. Specifically, Zhang et al. [15] reported that temperatures ranging from −179.8 °C to 99.6 °C reduced the compressive strength by 15%, down to 26.4 MPa, and the flexural strength by 49% to 2.9 MPa. These decreases were attributed to the formation of microcracks wider than 1 μm and increased porosity within the concrete matrix post-cryogenic exposure.

Micrometeorite impacts present another marked hazard in the lunar environment, distinct from extreme temperature variations. These naturally occurring interplanetary debris travel through space and frequently impact the lunar surface [16]. Unlike Earth, the Moon lacks an atmosphere to shield its surface from these high-velocity impacts. According to the Grun Lunar flux model, the frequency of these impacts reduces with increasing micrometeorite mass, revealing a higher incidence of impacts from smaller micrometeorites [17]. These particles hit the lunar surface at velocities ranging from 3 km/s to 70 km/s, with an average of 20 km/s [18]. Despite their small size, the high kinetic energy of micrometeorite impacts can lead to severe instantaneous damage, ranging from the degradation of the concrete to the catastrophic failure of structural components. In addition to micrometeorite scenarios, concrete structures are also frequently exposed to high-velocity impact and blast loading conditions, where the strain rate typically ranges from 10^2 to 10^4 s^{-1} [19]. To comprehensively evaluate the mechanical behaviour of concrete under extreme loading rates, it is essential to conduct dynamic tests, with Split Hopkinson pressure bar (SHPB) tests being particularly critical for replicating the high strain-rate shock waves and capturing the dynamic responses of concrete-like materials subjected to severe stress states [20–22]. Moreover, SHPB testing is vital for assessing the extreme pressure and strains that lunar concrete structures might endure due to explosive impacts and thermal shocks from rocket exhaust flames. Lian et al. [23] explored the dynamic performance of AAC with different strengths through SHPB tests using an Instron high-speed testing device. The study observed significant variations in dynamic increase factor (DIF) for compression (DIF_{fc}) at lower strain rates (10^{-5} ~ 350 s^{-1} for compression; 10^{-6} ~ 156 s^{-1} for split-tension) among different strengths. At higher strain rates, however, the variations in DIF_{fc} and the DIF for split-tension (DIF_{ft}) were comparatively minor. Furthermore, the empirical formulas developed by the Comité Euro-International du Béton (CEB) regarding strain rate effects on cement-based concrete were found to both overestimate and underestimate the predicted DIF_{fc} and DIF_{ft} values for AAC. Yao et al. [24] conducted a comprehensive study on the dynamic fracture toughness of slag-based alkali-activated mortars utilising notched semi-circular bend specimens. Their findings revealed that the fracture toughness increased markedly with the loading rate (up to 300 s^{-1}), curing time (up to 56 days), and moisture content. Wang et al. [25] performed high-temperature SHPB tests (up to 800 °C and 145 s^{-1}) on AAC specimens (Φ100 mm × 50 mm), indicating that at 200 °C, DIF_{fc} reached its peak value and the strain rate sensitivity of dynamic strength was most evident. Liu et al. [26] reported that the water cooling regime significantly compromised both the dynamic compressive strength and elastic modulus of the geopolymer concrete after thermal exposure ranging from 250 °C to 1000 °C. In terms of the lunar regolith simulant-based geopolymers, Wang et al. [27] examined the influence of alkali activator content and impact pressure using SHPB tests. They found that dynamic compressive strength enhanced dramatically with increasing strain rate between 40 s^{-1} and 110 s^{-1}. The peak stress was about 20 MPa at a 5% Na_2SiO_3 concentration, under which conditions the specimen showed minimal matrix damage. Moreover, a constitutive equation was developed based on a modified Zhu-Wang-Tang (ZWT) viscoelastic model, effectively capturing the combined effects of strain rate and dynamic responses of lunar geopolymers.

The comprehensive review conducted underscores the variability in both the static and dynamic properties of AAC under diverse environmental and stress scenarios. Despite prominent insights, investigation into the performance of lunar concrete, particularly under the extreme temperature conditions of the lunar environment, is conspicuously lacking. This study presents an extensive exploration of the temperature effects on the mechanical behaviour of a novel alkali-activated lunar high-strength concrete (LHSC). Critical factors including the testing temperature (20 °C, −70 °C, and −170 °C) and strain rate (50–200 s^{-1} for compression and 30–90 s^{-1} for split-tension) were evaluated to determine the static and dynamic characteristics of LHSC. The findings from this investigation notably contribute to filling the existing research gap and ensuring the material and structural safety, thereby supporting the sustainable human habitation on the Moon.

2. Experimental setup and methodology

2.1. Materials and mixture proportion

To enhance the fidelity of replicating the lunar surface soil environment while optimizing cost-effectiveness and resource utilisation, a commercial lunar regolith simulant designated as Lunar Highlands Simulant (LHS-1) was used. This simulant meticulously captures the texture of lunar soil by integrating mineral and rock fragments in precise proportions derived from analyses of regolith samples returned by the Apollo missions [28]. Its particle sizes extend from 0.04 to 1000 μm, and the median particle size measures 51 μm. Given the chemical composition and fineness of LHS-1 lead to its low reactivity in alkaline environments [29,30], supplementary terrestrial materials such as Class F fly ash, silica fume, and ground granulated blast-furnace slag (GGBFS) were added to the mixes to enhance the reactivity of the precursors and improve the performance of the resulting lunar geopolymers. Table 1 lists the oxide compositions and physical properties of these powdery materials. Regarding the alkali activator, a pre-blended solution consisting of approximately 14 wt% sodium hydroxide (NaOH) pellets and 86 wt% sodium silicate (Na_2SiO_3) solution was utilised to create a highly alkaline environment essential for initiating the chemical activation of precursors. Table 2 details the mix ratio of LHSC used in this study, selected as the optimal formulation based on its mechanical properties and compatibility for lunar applications, as concluded in prior research [31]. It is important to note that the reported water proportion in this mix refers exclusively to the externally added water and does not account for the intrinsic water present in the Na_2SiO_3 component of the activator solution.

2.2. Specimen preparation and curing

During the preparation of the specimens, all powder constituents

Table 1
Chemical constituents and physical characteristics for LHSC precursor materials.

Precursors	Oxide compositions (wt%)					SG	Density (kg/m^3)	Fineness (m^2/kg)	LOI (%)	PSD (μm)
	CaO	SiO_2	Al_2O_3	Fe_2O_3	MgO					
LHS−1	13.5	49.1	26.3	3.2	2.9	2.75	1270	–	0.41	0.04–1000
GGBFS	43.6	35.4	10.8	0.4	8.1	2.81	1275	395	0.48	0.2–100
Fly ash	1.1	67.9	23.8	6.3	0.2	2.42	1041	430	1.20	0.3–110
Silica fume	1.9	89.6	2.1	1.5	0.8	2.23	625	16000	3.80	0.03–0.2

Note: "SG" denotes specific gravity, "LOI" denotes loss on ignition, "PSD" denotes particle size distribution, and "–" denotes unreferred parameter.

Table 2
Relative proportions of precursors and water to activator in LHSC.

LHS-1	GGBFS	Silica fume	Fly ash	Activator	Water
2.38	1.26	0.13	0.20	1.00	0.13

were initially subjected to dry mixing for a duration of 3–5 min. Subsequently, the alkali activator and water were gradually introduced into the blend. This wet mixing phase was extended for an additional 3–5 min to ensure effective activation of the precursor materials and to achieve a homogenous mixture. The resultant mixtures were then poured into polyvinyl chloride (PVC) pipe moulds with an inner diameter of 50 mm. To improve the consolidation of the mixture and minimise entrapped air, a vibration table was employed during the casting process. Following the initial setting phase, the specimens were subjected to a curing process within an oven, maintained at 96.7 °C for 48 h. The adopted thermal protocol was specifically designed to simulate the thermal characteristics of a lunar day, which spans approximately 324–372 h at a latitude of 30 degrees [32]. The vacuum condition on the Moon was not simulated in the curing process of the LHSC, following the findings of Zhou et al. [32] who reported no negative effects on lunar geopolymers in non-vacuum environments. Fig. 1 shows the LHSC following casting and prior to testing, where the cylindrical specimens were sectioned into multiple 25 mm segments using a high-speed cutting saw in preparation for subsequent mechanical performance evaluations.

2.3. Testing methods

2.3.1. Temperature setup

In this study, the mechanical performance of LHSC was evaluated at ambient (approximately 20 °C) and cryogenic temperatures (approximately −70 °C and −170 °C). These temperatures were selected to represent both terrestrial baseline conditions (20 °C) and the extreme thermal environments encountered on the lunar surface. The lowest temperature (-170 °C) approximates the severe cold experienced during the lunar night or within permanently shadowed regions near the lunar poles, where surface temperatures can reach as low as −173 °C [33]. The intermediate temperature (-70 °C) represents subzero thermal conditions that may arise in thermally buffered environments, such as partially shieled and subsurface structures, or during transitional thermal phases. For cryogenic testing, specimens were conditioned in a specialized cryogenic chamber (380 mm × 230 mm × 150 mm), as shown in Fig. 2. The chamber was insulated in a polyurethane box to minimise the external thermal effects. Controlled cooling was achieved by incrementally introducing the liquid nitrogen from a Dewar vessel at a rate of 1 °C/min. Internal temperatures within the chamber were continuously monitored and displayed on a screen linked to the internal temperature sensor. Once the target temperatures were reached, the specimens were stabilized at these conditions for an additional two hours to achieve uniform thermal equilibration [34]. Then, the specimens were immediately transferred to the loading apparatus for mechanical testing.

2.3.2. Quasi-static test

Quasi-static mechanical behaviour evaluations of LHSC specimens were carried out under the established temperature conditions utilising uniaxial compressive and split-tensile tests. These tests used cylindrical concrete specimens (Ø50 × 25 mm) to facilitate accurate determination of the quasi-static strength, which is crucial for calculating the DIF for high strain rate conditions. Compressive tests were conducted employing a servo-hydraulic testing machine with a capacity of 500 kN and a loading rate of 1 kN/s, as per ASTM C39 [35]. Split-tensile tests were performed at a loading rate of 0.05 kN/s according to ASTM C496 [36], with wooden strips (30 mm × 5 mm × 2 mm) placed on the top and bottom of the specimens to reduce stress concentrations. Details of the testing setup are illustrated in Fig. 3.

2.3.3. Dynamic impact test

The SHPB test is a fundamental technique used to evaluate the dynamic responses of materials under high strain rates. The setup includes essential components such as the striker, incident, and transmission bars, with the specimen positioned between the latter two bars. SHPB test operates on the principle of stress wave propagation in the elastic bars. As stipulated by the stress wave theory, the stress and strain in the specimen adhere to the premises set forth by the three-wave theory method in SHPB tests [37]. Upon the striker bar impacting the incident bar, a stress wave is generated that propagates through the bar system. Strain data are recorded using two resistance strain gauges affixed to both the incident and transmission bars, capturing incident (ε_i),

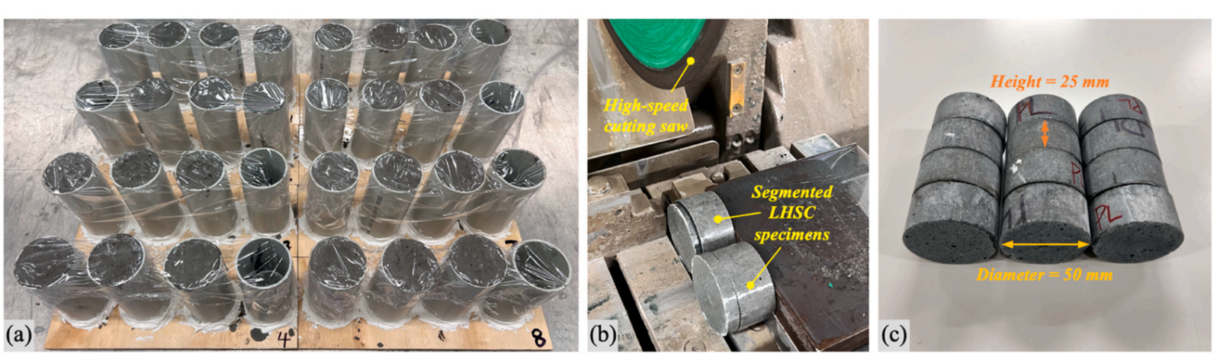

Fig. 1. (a) LHSC mixtures casted in PVC moulds and (b-c) segmented specimens prior to testing.

Fig. 2. Temperature setup for LHSC specimens under cryogenic conditions.

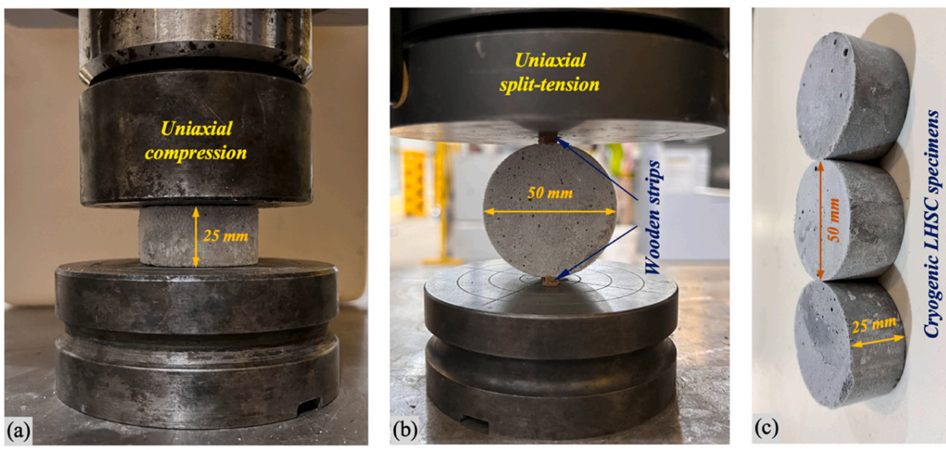

Fig. 3. Quasi-static testing setup for uniaxial (a) compression and (b) split-tension using (c) cryogenic cylindrical LHSC specimens.

reflected (ε_r), and transmitted (ε_t) strain signals. The measurements facilitate precise calculations of the dynamic stress, strain, and strain rate experienced by specimens. The equations governing these calculations, derived from stress wave theory, are expressed as follows [23,38]:

$$\sigma(t) = EA \times \varepsilon_t(t) \; / \; A_s \tag{1}$$

$$\varepsilon(t) = -2C \times \int_0^t \varepsilon_r(t)dt \Big/ L_s \tag{2}$$

$$\dot{\varepsilon}(t) = -2C \times \varepsilon_r(t) \; / \; L_s \tag{3}$$

where $\sigma(t)$, $\varepsilon(t)$, and $\dot{\varepsilon}(t)$ represent the stress, strain, and strain rate experienced by specimens at interfaces with the loading bar, respectively. E represents the elastic modulus for bar materials, A and A_s denote the cross-sectional areas of the bars and specimens, respectively, C is the wave speed in the bars, and L_s is the specimen length.

This study encompassed both compressive and split-tensile impact tests conducted on cylindrical LHSC specimens (Ø50 × 25 mm), utilising an SHPB apparatus with bar diameters of 100 mm. The striker, incident, and transmission bars measured 500 mm, 4500 mm, and 3000 mm in length, respectively, with a density of 7.85 g/cm^3. SHPB tests were performed at various strain rates (50 s^{-1}, 100 s^{-1}, 150 s^{-1}, and 200 s^{-1} for compression; 30 s^{-1}, 50 s^{-1}, 70 s^{-1}, and 90 s^{-1} for split-tension) under three different temperatures (20 °C, −70 °C, and −170 °C). The strain rates were precisely controlled by adjusting the impact speed of the striker bar. These strain rate ranges fall within the commonly used intervals for SHPB testing of concrete materials, ensuring relevance to high-velocity impact and blast loading scenarios [19,39,40]. To mitigate the fundamental issues of significant pulse oscillation and dispersion characteristic in SHPB impact, a Ø20 × 1 mm rubber sheet positioning between the striker and incident bars was employed as a waveform shaper. Due to the pronounced diameter mismatch between the specimen and bars, a high-sensitivity semi--conductor strain gauge was mounted on the transmission bar to accurately detect the subtle strain signals during wave propagation. Furthermore, the failure evolution of the LHSC specimens was documented using a high-speed camera operating at a frame rate of 5 million frames per second (fps), enabling the precise observation of crack initiation and propagation under dynamic loading. Fig. 4 shows the experimental setup for both compressive and split-tensile SHPB impact testing of LHSC specimens, while Fig. 5 provides a flowchart outlining the procedures for quasi-static and dynamic mechanical tests under various temperature conditions.

It should be pointed out that in the dynamic split-tensile tests, the specimen ends were directly in contact with the incident and transmission bars without the use of wooden bearing strips. For short cylindrical discs, this configuration generates a nearly uniform diametral tensile stress field comparable to that of quasi-static split-tensile tests, as validated by the analytical and numerical studies [41–43].

3. Results and discussions

3.1. Quasi-static mechanical behaviour

The quasi-static mechanical tests were carried out on LHSC specimens exposed to temperatures ranging from 20 °C to −170 °C, with the results depicted in Fig. 6. The data reveal that both the compressive and split-tensile strengths are notably influenced by the matrix temperature during testing. Specifically, the compressive strength of LHSC enhanced progressively from 129.5 MPa at 20 °C to 144.2 MPa at −70 °C and

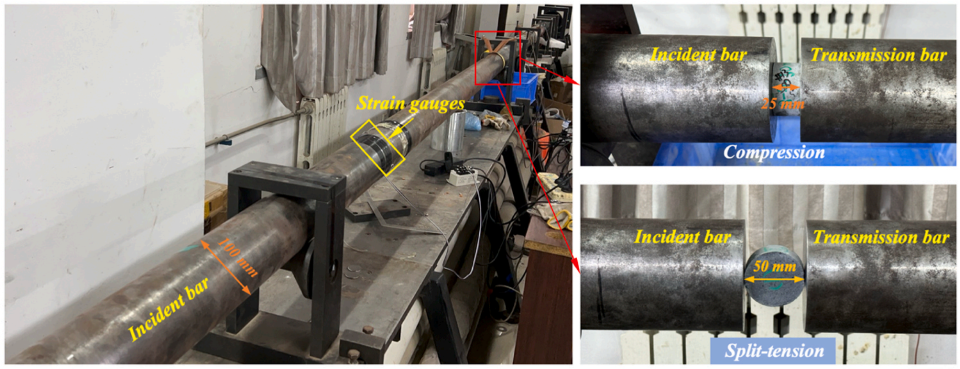

Fig. 4. SHPB test setup for dynamic compression and split-tension on LHSC specimens.

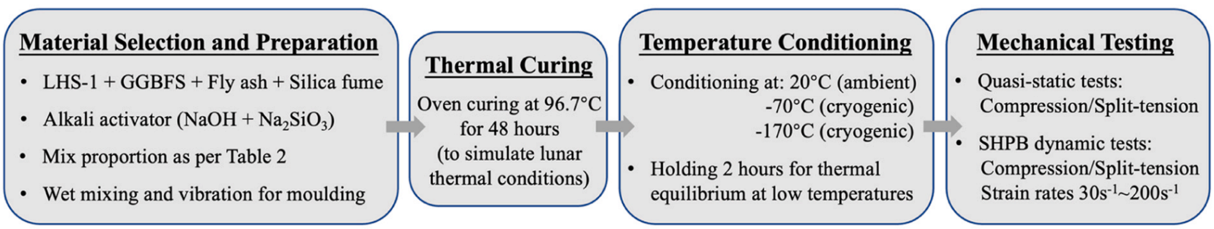

Fig. 5. Flowchart for mechanical testing of LHSC under varying temperature conditions.

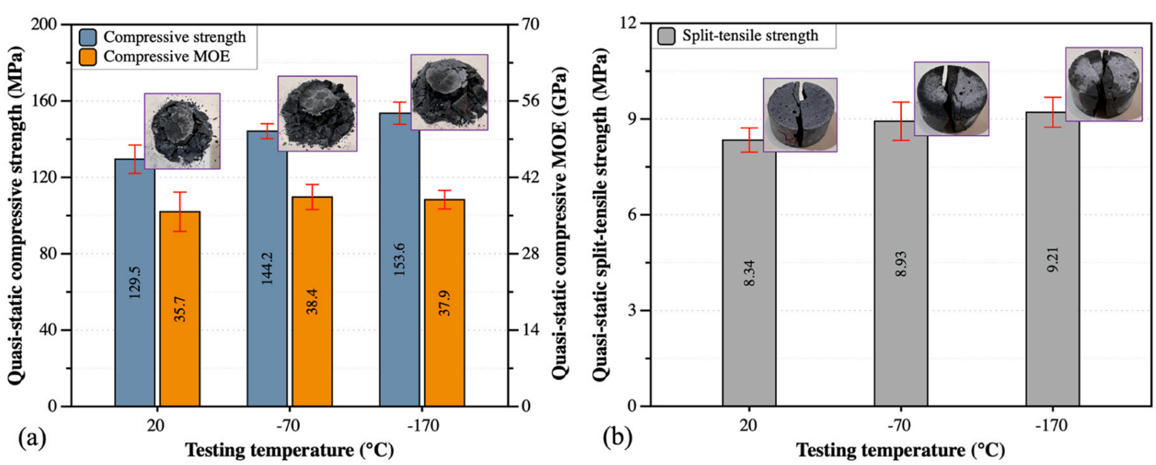

Fig. 6. Quasi-static (a) compressive and (b) split-tensile performance of LHSC specimens.

further to 153.6 MPa at −170 °C, corresponding to increases of 11.4% and 18.6%, respectively. Similarly, the split-tensile strength increased from 8.3 MPa to 8.9 MPa and then to 9.2 MPa, marking increases of 7.2% and 10.8% under the same thermal conditions. This property can be ascribed to the freezing of pore water within the concrete matrix into ice crystals with higher stiffness and strength [44]. These crystals fill the micro-pores in the matrix, enhancing the microstructure and contributing to an increased overall stiffness [45]. The compressive modulus of elastic (MOE), as shown in Fig. 6(a), therefore exhibited a rising trend from 35.7 GPa to 38.4 GPa under the studied temperature conditions. The marked reduction in ambient temperature significantly boosts the LHSC resistance to compressive deformation and final failure, as evidenced by the more intact core and the larger crushed pieces of the specimen after compressive testing. However, the increase in stiffness due to the evident temperature drop led the LHSC to exhibit a more pronounced brittle failure when subjected to split-tensile loads, as depicted in Fig. 6(b).

3.2. Dynamic impact behaviour in compression

Table 3 shows the results of the dynamic compression testing at different temperatures, including parameters such as strain rate, dynamic compressive strength, MOE, and DIF across all LHSC specimens. A detailed examination of these results, along with the failure modes observed in the impacted specimens, will be elaborated in subsequent sections.

3.2.1. Dynamic compressive strength and stress-strain relationship

Fig. 7 shows the dynamic compressive strength variations of LHSC under temperature-dynamic coupled conditions. The data delineated a marked linear increase in dynamic strength with rising strain rates across all tested temperatures, as evidenced by the purple fitting curves ($R^2 \approx 0.96$–0.98). A notable trend observed was the steepening of these curves as the temperature decreased from 20 °C to −70 °C and −170 °C. This behaviour may be due to the strengthening impact of ice crystals formation inside the geopolymer at extreme temperatures. With

Table 3
Summary of dynamic compression results of LHSC at different temperatures.

Testing temperature (°C)	Strain rate (s^{-1})	Dynamic compressive strength (MPa)	Dynamic MOE (GPa)	Compressive DIF
Ambient 20	48.29	148.6	38.2	1.147
	50.53	148.2	37.4	1.144
	60.87	153.8	39.7	1.188
	101.26	162.7	41.6	1.257
	102.76	160.3	42.3	1.238
	119.11	172.8	43.6	1.334
	124.84	176.4	43.0	1.362
	144.18	183.4	45.5	1.417
	148.11	186.6	44.9	1.441
	151.30	185.2	43.8	1.430
	169.89	192.3	44.1	1.485
	191.10	197.2	45.2	1.523
	204.01	200.1	45.5	1.545
	205.16	197.0	46.4	1.521
Cryogenic −70	53.46	167.2	40.5	1.160
	57.92	175.1	41.7	1.214
	86.54	186.4	43.9	1.292
	98.83	199.5	43.6	1.384
	105.55	197.5	45.3	1.370
	127.62	204.0	47.0	1.415
	149.86	223.0	50.4	1.546
	151.24	218.2	49.8	1.513
	178.66	226.7	49.0	1.572
	201.47	235.5	48.4	1.633
Cryogenic −170	54.28	189.6	41.8	1.235
	59.14	187.2	40.9	1.219
	96.49	205.8	43.5	1.340
	101.95	213.3	44.8	1.389
	110.53	220.9	45.6	1.438
	143.39	239.8	46.7	1.561
	147.82	235.0	46.1	1.530
	152.36	242.4	45.9	1.578
	201.74	262.4	44.7	1.708
	202.33	257.1	44.1	1.674

temperatures lowering to −173 °C, ice crystals experienced a notable increase in the peak stress [46]. This enhancement effectively reinforced the LHSC microstructure by filling pores and impeding crack propagation, thereby improving the matrix integrity under cryogenic conditions. It is noteworthy that the load-bearing capacity of the concrete matrix is closely related to its cohesive strength. The formation of ice increased the matrix stiffness and bridging across pores despite the cohesive energy of the matrix may change with temperature and strain rate [47]. Specifically, at 20 °C, as the strain rate increased from approximately 50 s^{-1} to 200 s^{-1}, the dynamic compressive strength enhanced from about 148 MPa to 200 MPa, representing an increase of 14.4–54.5% relative to its baseline static strength of 129.5 MPa (refer to Fig. 6(a)). At −70 °C, maintaining consistent strain rates, the strength rose from approximately 167 MPa to 235 MPa, exceeding the static value of 144.2 MPa by 16.0–63.3%. At the extreme temperature of −170 °C, the dynamic strength values ranged from 189 to 262 MPa, surpassing the static strength of 153.6 MPa by 23.4–70.8%. The transformation of pore water into ice crystals within the geopolymer's porous structure provides a denser filling and works in conjunction with rate hardening effect to delay crack coalescence and thus improve the load-carrying capability of LHSC.

Fig. 7 likewise illustrates the variations in dynamic compressive MOE of LHSC under different testing conditions. The MOE values were calculated from the initial linear slope of the dynamic stress-strain curves (see Fig. 8), specifically within the strain range of 0–0.002, to ensure that the modulus reflects the elastic response prior to yielding. At an ambient temperature of 20 °C, the dynamic MOE, represented by yellow markers, showed a clear upward trend with increasing strain rate, as shown by the green fitting line ($R^2 = 0.879$). Specifically, the MOE increased from 37.4 GPa at a strain rate of 50.53 s^{-1} to 46.4 GPa at 205.16 s^{-1}, reflecting an enhancement of 4.8–30.0% relative to the static MOE of 35.7 GPa, as shown in Fig. 6(a). As temperatures decreased to −70 °C, the MOE at similar strain rates not only increased but also demonstrated a steeper sensitivity to strain rate changes. As an illustration, at 105.55 s^{-1}, the MOE reached 45.3 GPa, up by 18.0% from its quasi-static counterparts. However, beyond a strain rate of around 149 s^{-1}, the increase in MOE began to plateau, peaking at a 31.3% improvement and then reducing to 26.0% at 201.47 s^{-1}. At −170 °C, while the initial trend similarly showed an increase in MOE with strain rate, it plateaus sooner and at some points decreased under higher strain rates, demonstrating a reduced enhancement effect in extreme cold conditions. For instance, the greatest improvement was about 23.2% at a strain rate near 143 s^{-1}, but it diminished to 16.4% by 202.33 s^{-1}. This data suggests that a reduction in temperature from 20 °C to −70 °C markedly augments the strain rate-induced increase in dynamic MOE. However, further lowering temperatures to −170 °C does not consistently enhance MOE values, especially beyond certain strain rate thresholds. This can be attributed to the rapid formation and propagation of microcracks, resulting from the volumetric expansion of freezing pore water, which led to a less cohesive concrete matrix.

Fig. 8 shows the typical dynamic compressive stress-strain relationships of LHSC under varying strain rates and temperature conditions, plotted according to the formulae exhibited in Eqs. (1)-(3). The data underscored a distinct sensitivity to both strain rate and testing temperature. An increase in strain rate coupled with a decrease in temperature resulted in a substantial elevation in peak stress. Notably, at elevated strain rates, the initial slope of the curve steepened evidently,

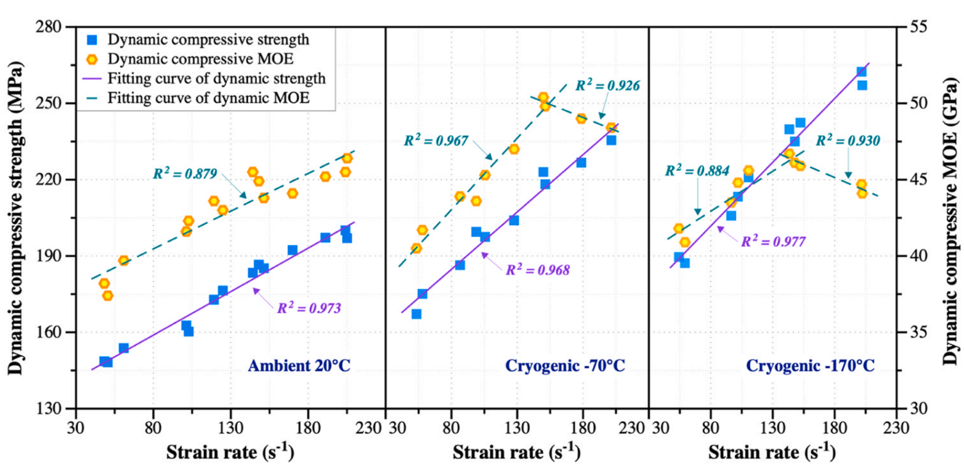

Fig. 7. Dynamic compressive strength and MOE of LHSC under different temperatures and strain rates.

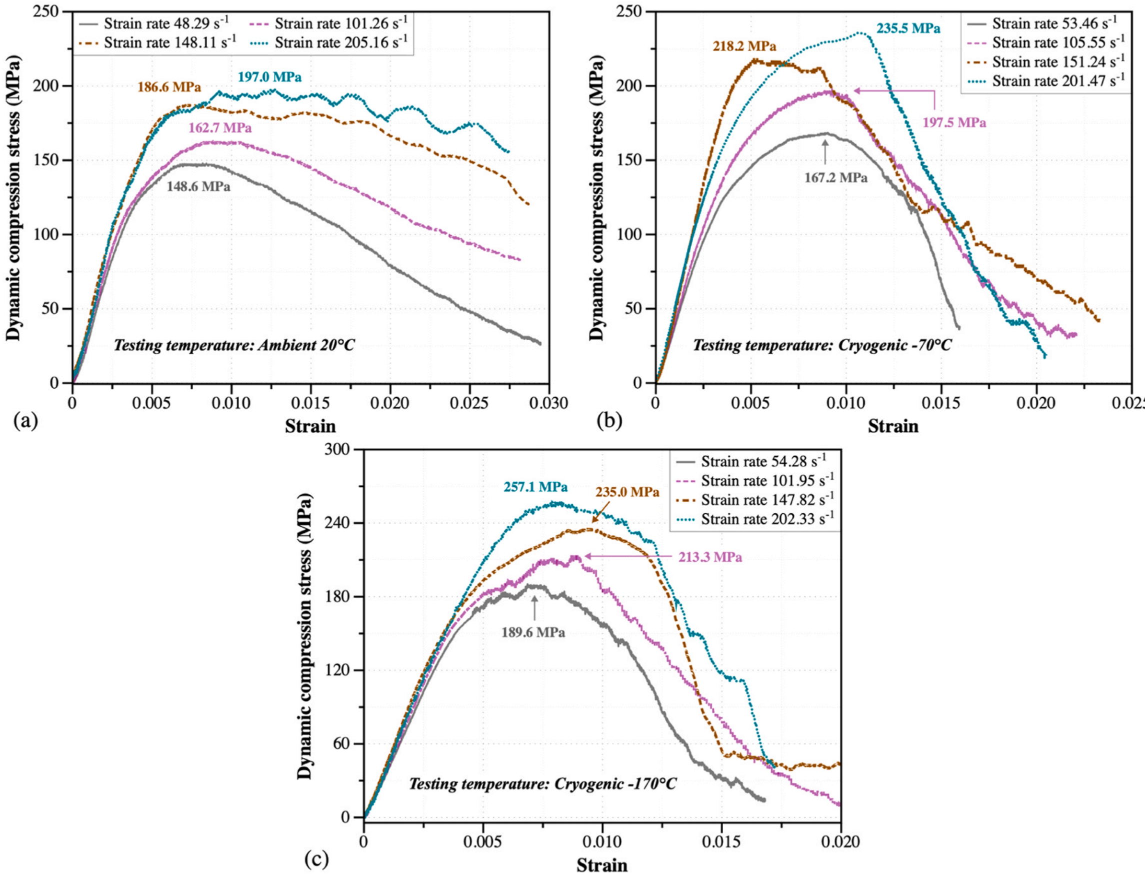

Fig. 8. Dynamic compressive stress-strain relationship of LHSC at (a) 20 °C, (b) −70 °C, and (c) −170 °C.

as observed at 20 °C and −70 °C, indicating an intensified strain rate hardening effect. In addition, a pronounced temperature decrease led to marked post-peak softening, highlighting a diminished ability of the LHSC matrix to tolerate deformation. For example, at strain rates of approximate 100 s^{-1} and a strain of 0.015, the residual load-carrying capacity post-peak at 20 °C, −70 °C, and −170 °C equated to 89.1%, 32.6%, and 30.5%, respectively, of their peak stresses. This behaviour is attributable to the inherently brittle nature of the geopolymer, which intensified under extreme cold due to the freezing of pore water [44,48]. The formation of ice significantly increased both the stiffness and load-carrying capacity of the lunar geopolymer. However, this also reduced the matrix ability to resist deformation, resulting in a more abrupt decline in the stress post-peak at cryogenic temperatures and higher strain rates.

3.2.2. Energy absorption capability

The energy absorption capability, W, is crucial for investigating the deformation and toughness performance of the LHSC. It is determined by the ductility and ultimate strength of the material, with higher absorption indicating better resistance to dynamic loadings [49]. This behaviour is assessed via calculating areas beneath the stress-strain curve under dynamic compression, which illustrates the total energy dissipated as the concrete deforms and fails, given by [50]:

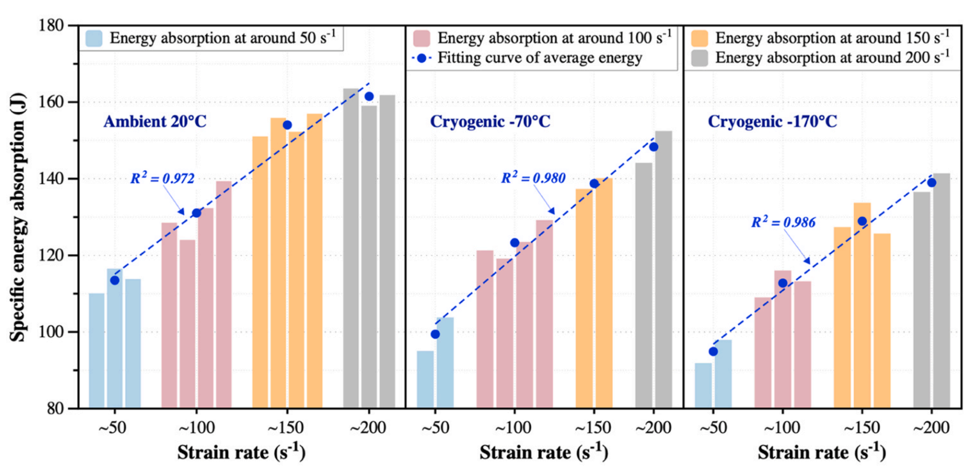

Fig. 9. Compressive energy absorption of LHSC across various temperatures and strain rates.

$$W = \int_0^{\varepsilon} \sigma(\varepsilon)\mathrm{d}\varepsilon \tag{4}$$

where $\sigma(\varepsilon)$ represents the stress at a specific strain ε. Fig. 9 illustrates the variations in energy absorption of LHSC, derived from the curves' strain reaching 0.020 under different conditions. Observations showed a linear increase in energy absorption values with strain rate at all testing temperatures. As an illustration, at ambient temperature, the average values increased from 113.5 J to 161.5 J as the strain rate rose from around 50 s^{-1} to 200 s^{-1}, an increment of 42.3 %. However, the absorption ability of LHSC reduced to a certain extent under cryogenic conditions, dropping to 148.3 J and 139.0 J at approximately 200 s^{-1} for −70 °C and −170 °C, respectively. This decline highlights the adverse impacts of extreme cold on the toughness of lunar geopolymers.

The residual load-carrying capacity of the LHSC under cryogenic temperatures was significantly reduced, especially once the strain exceeded 0.020, indicating pronounced strain softening post-peak, as observed in Fig. 8. In contrast, at 20 °C, LHSC retained significant residual load-bearing capacity at various strain rates even beyond a 0.020 strain, suggesting that its dynamic absorption capacity at ambient temperature was substantially higher than previously calculated. That is, the influence of cryogenic conditions on the dynamic toughness was notably pronounced. This effect may stem from the transformation of pore water into ice under cryogenic conditions, filling voids inside the lunar geopolymer, signally increasing its stiffness. This conversion can cause volume expansion, potentially leading to microcracking and structural alterations [51]. Additionally, the differential thermal contraction and phase transitions between the ice and the matrix binder contributed to the freeze-induced microcrack defects within the geopolymers [52]. These factors collectively attenuated the capability of LHSC to absorb energy under impacts.

3.2.3. Dynamic failure mode and specimen impact process

Fig. 10 displays the failure pattern of LHSC specimens after the dynamic compressive test under various temperature and strain rate conditions. As the strain rate rises at a constant temperature, there is a marked increase in the severity of fragmentation, resulting in finer debris. For instance, at 20 °C, specimens subjected to lower strain rates yielded larger, block-like fragments, whereas those exposed to higher strain rates (above 102.8 s^{-1}) produced smaller, more fragmented pieces. Conversely, at the similar strain rate, lowering temperatures to cryogenic conditions significantly increased the matrix brittleness. This change was apparent as large block-like fragments existed even at strain rates around 150 s^{-1}. Moreover, at strain rates near 200 s^{-1}, despite severe crushing, a significant quantity of angular debris remained embedded within the residual matrix. Notably, at lower strain rates below 100 s^{-1}, particularly at −170 °C and 49.1 s^{-1}, the specimens exhibited minimal matrix damage and largely retained their original geometry. These observations further indicated that lower testing temperatures significantly enhanced the brittleness of LHSC. High strain rates expedited crack propagation, leading to abrupt fracturing. Conversely, ice formation in the matrix at cryogenic temperatures served as additional reinforcements, potentially delaying cracks initiation and expansion [45], thereby enhancing the LHSC resistance to deformation and damage under dynamic loading conditions.

To enhance the understanding of the temporal development of damage leading to ultimate failure patterns, high-speed photographic sequences were meticulously analysed subsequent to dynamic failure mode assessment. Fig. 11 presents these sequences depicting typical LHSC specimens subjected to a strain rate of approximately 150 s^{-1} at three distinct temperatures: 20 °C, −70 °C, and −170 °C. Irrespective of temperature conditions, a consistent failure process was observable, characterized by crack initiation, propagation, and final fragmentation beyond 240 μs. At each specified temperature, an increase in time

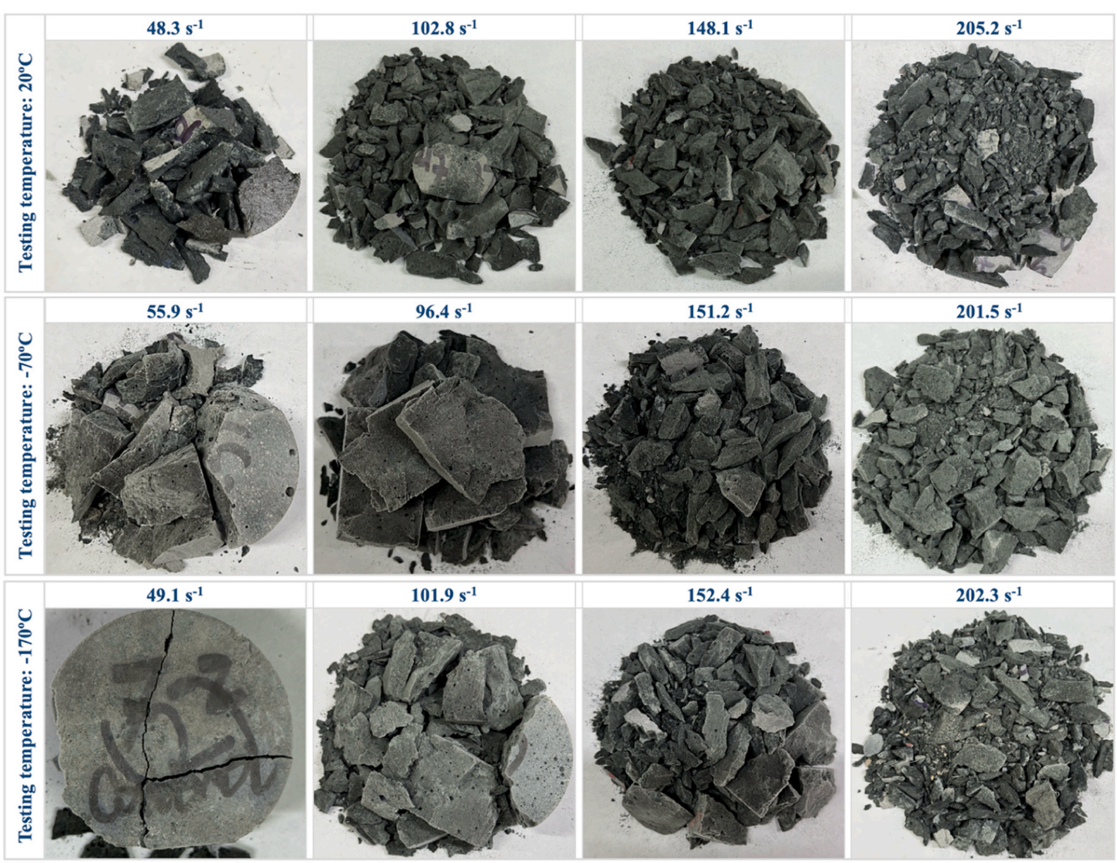

Fig. 10. Compressive failure patterns in LHSC specimens subjected to SHPB test at variable temperatures.

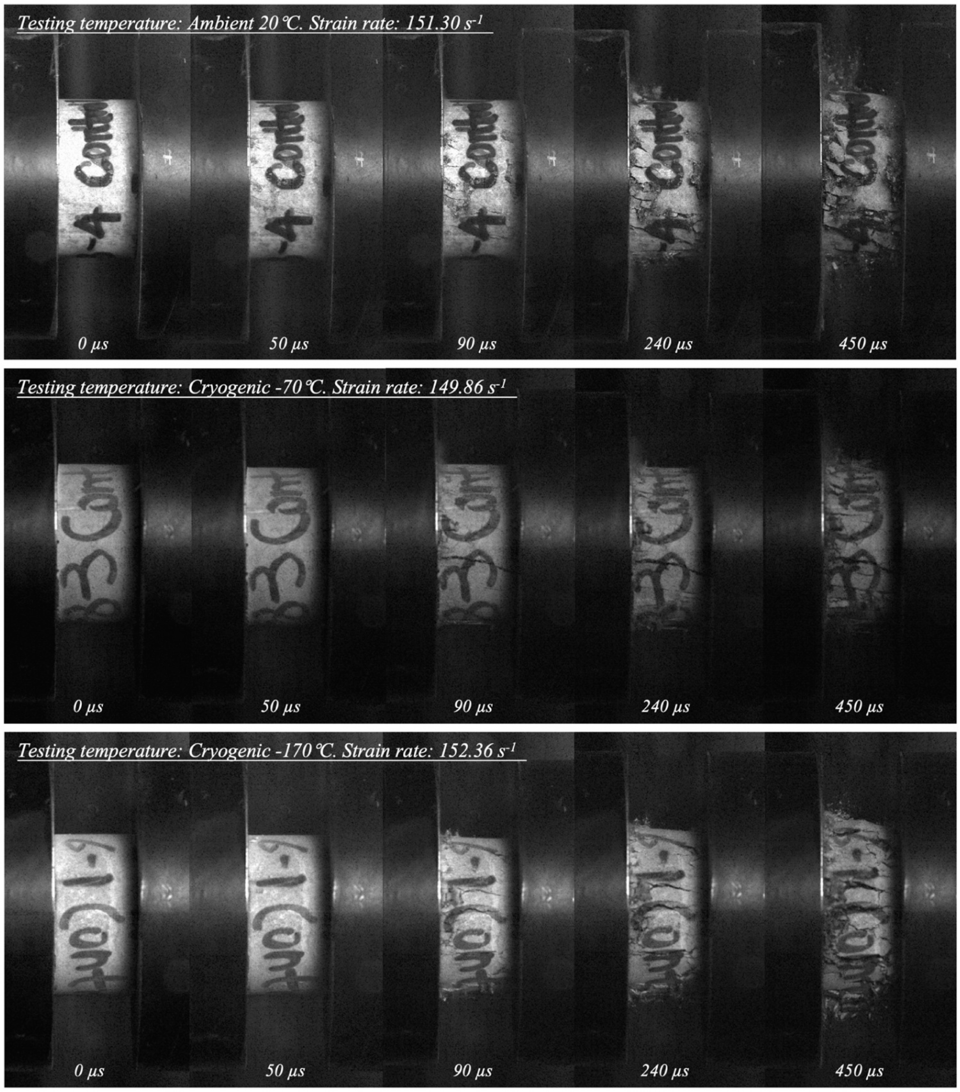

Fig. 11. Compressive failure process for LHSC under various temperatures.

correlated with more discernible fracture patterns and crack coalescence, consistent with debris morphologies and fragment characteristics previously discussed.

Comparative analysis of the identical timeframes under varying temperatures revealed several distinctive differences. At 50 μs, the specimen at the ambient 20 °C exhibited initial microcracks on its surface, while specimens at −70 °C and −170 °C showed no visible microcracks, indicating a delayed fracture onset under cryogenic conditions. By 90 μs, the microcracks at 20 °C had expanded, and longitudinal microcracks commenced formation near the specimen-transmission bar interface. In stark contrast, at −70 °C and −170 °C, pronounced longitudinal microcracks had penetrated the surface, with the −170 °C specimen additionally displaying surface debris spalling, signifying increased brittleness. By 240 μs, the specimen at the ambient condition demonstrated mild spalling and a few penetrating microcracks, whereas under cryogenic conditions, both the quantity and width of these cracks markedly expanded, accompanied by more substantial debris spalling. By 450 μs, the −70 °C specimen exhibited less severe matrix fracturing as compared to the other two temperature specimens, which underwent extensive fragmentation.

These findings underscored that cryogenic temperatures, particularly −170 °C, not only delayed initial crack formation but also accelerated subsequent crack propagation, resulting in more rapid and widespread failure once the fracture process was initiated. According to Zhang et al. [53], temperatures below −90 °C altered the pore structure, either by collapsing or filling larger voids and concurrently increasing the proportion of smaller pores. The elimination of these larger pores notably diminished the concrete capacity to accommodate deformation, thereby intensifying its brittleness and hastening crack propagation once initiated.

3.2.4. *Strain rate sensitivity*

DIF is widely employed to quantify the strain rate effect on concrete-like materials by comparing their dynamic strength to static strength. This parameter is crucial for evaluating the material's enhanced resistance under impact or shock conditions. Notable studies, such as those by CEB-FIP [54], Bischoff et al. [55], and Malvar et al. [56,57], have proposed empirical formulas based on the rate-dependent behaviour of ordinary concrete. Particularly, CEB-FIP has recommended a two-part relationship between DIF_{fc} and strain rates, serving as references for global research on concrete materials.

Fig. 12 depicts the results of DIF_{fc} for LHSC tested at various strain rates across three different temperatures. The results demonstrated a consistent increase in DIF_{fc} with rising strain rate at all tested temperatures. Notably, the sensitivity to strain rate, as represented by the slope of the data fitting curves, became more prominent with decreasing

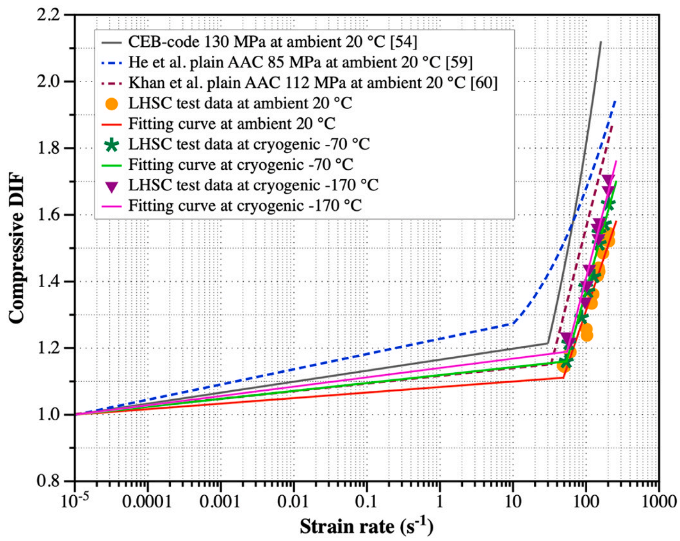

Fig. 12. Relationship between DIF_{fc} and strain rate of LHSC across various temperatures.

temperatures from ambient to cryogenic conditions. For example, at an ambient temperature of 20 °C, the DIF_{fc} values increased from 1.144 to 1.545 as the strain rate increased from approximately 50 s^{-1} to 200 s^{-1}, marking a 35.1% rise. At −70 °C and −170 °C, within the same strain rate range, DIF_{fc} values increased from 1.160 to 1.633 (a 40.8% rise) and from 1.219 to 1.708 (a 40.1% rise), respectively. Moreover, at the specific strain rate, as an illustration, around 150 s^{-1}, the DIF_{fc} values at cryogenic temperatures showed enhancements of 5.8% and 10.4%, respectively, as compared to ambient conditions. The underlying mechanism may be due to the freezing of pore-water at ultra-low temperatures, which formed ice networks, naturally bridging the micro-cracks and voids [58]. This process enhanced the bonding between aggregates and the binder while decreasing the overall deformability of the geopolymer matrix. Such synergistic effects led to a more pronounced rate-hardening behaviour once the strain rate exceeded a transition threshold, amplifying the rate-dependent strengthening of the LHSC at cryogenic temperatures.

The fitted equations for DIF_{fc} of the current LHSC specimens across different temperatures are detailed in Eqs. (5)-(7). The transition strain rates were identified as approximately 48.29 s^{-1} at ambient 20 °C, 53.46 s^{-1} at −70 °C, and 54.28 s^{-1} at −170 °C. Fig. 12 also juxtaposes the CEB-FIP model curve for 130 MPa AAC and two DIF_{fc} curves for plain AAC at 20 °C as proposed by He et al. [59] and Khan et al. [60]. As shown, the CEB-FIP model tended to overpredict the dynamic behaviour of LHSC at ambient temperature. Additionally, this study-derived DIF_{fc} curve at 20 °C demonstrated less rate sensitivity and lower DIF_{fc} values at given strain rates compared to those reported for plain AAC at the same temperature. This discrepancy can be attributed to the denser microstructure of LHSC prepared in this study, which delivered a higher uniaxial compressive strength (approximately 130 MPa, compared to 85 and 112 MPa for the other AACs), resulting in reduced strain rate sensitivity under dynamic loading [39].

$$DIF_{fc}\ \text{at ambient } 20\ ^\circ C = \begin{cases} 0.017\log\dot{\varepsilon} + 1.083 & \dot{\varepsilon} \le 48.29s^{-1} \\ 0.649\log\dot{\varepsilon} + 0.014 & \dot{\varepsilon} > 48.29s^{-1} \end{cases} R^2 = 0.939 \tag{5}$$

$$DIF_{fc}\ \text{at cryogenic} - 70\ ^\circ C = \begin{cases} 0.024\log\dot{\varepsilon} + 1.119 & \dot{\varepsilon} \le 53.46s^{-1} \\ 0.794\log\dot{\varepsilon} - 0.214 & \dot{\varepsilon} > 53.46s^{-1} \end{cases} R^2 = 0.974 \tag{6}$$

$$DIF_{fc}\ \text{at cryogenic} - 170\ ^\circ C = \begin{cases} 0.028\log\dot{\varepsilon} + 1.140 & \dot{\varepsilon} \le 54.28s^{-1} \\ 0.849\log\dot{\varepsilon} - 0.287 & \dot{\varepsilon} > 54.28s^{-1} \end{cases} R^2 = 0.965 \tag{7}$$

3.3. Dynamic impact behaviour in split-tension

Table 4 presents the experimental data from SHPB tests, detailing the strain rate, dynamic split-tensile strength, and DIF for LHSC specimens under varying temperature conditions. A thorough discussion of these results, alongside an analysis of the failure patterns observed in the impacted specimens, will be provided in subsequent sections.

3.3.1. Dynamic split-tensile strength

Fig. 13 illustrates the dynamic splitting-tensile strength of the LHSC tested under strain rates of approximately 30 s^{-1}, 50 s^{-1}, 70 s^{-1}, and 90 s^{-1} at different temperatures. The strength consistently exhibited a linear upward trend with increasing strain rate, as indicated by the fitting equations for each temperature group. For example, at 20 °C, the fitting curve ($y = 0.109x + 8.561$, $R^2 = 0.987$) showed a noticeable increase in split-tensile strength when the strain rate increased from

Table 4
Summary of dynamic split-tension results of LHSC at different temperatures.

Testing temperature (°C)	Strain rate (s^{-1})	Dynamic split-tensile strength (MPa)	Split-tensile DIF
Ambient 20	27.27	11.94	1.432
	30.07	11.84	1.419
	32.38	12.23	1.467
	50.51	13.94	1.672
	51.66	13.58	1.628
	54.32	14.21	1.704
	69.15	16.17	1.939
	69.33	15.93	1.910
	72.63	16.75	2.009
	89.62	18.39	2.205
	92.31	19.03	2.282
	94.48	18.60	2.231
Cryogenic −70	29.67	16.68	1.867
	33.58	17.38	1.946
	45.86	19.61	2.197
	49.28	19.22	2.152
	65.83	21.78	2.439
	68.24	21.26	2.381
	74.15	22.14	2.479
	86.35	23.81	2.666
	87.92	24.29	2.720
	92.61	23.66	2.649
Cryogenic −170	28.23	20.61	2.237
	29.84	19.98	2.169
	46.36	21.62	2.348
	47.62	21.32	2.315
	60.60	23.41	2.542
	66.09	24.06	2.612
	73.21	23.60	2.562
	89.64	25.84	2.806
	95.66	26.06	2.829
	98.48	25.55	2.774

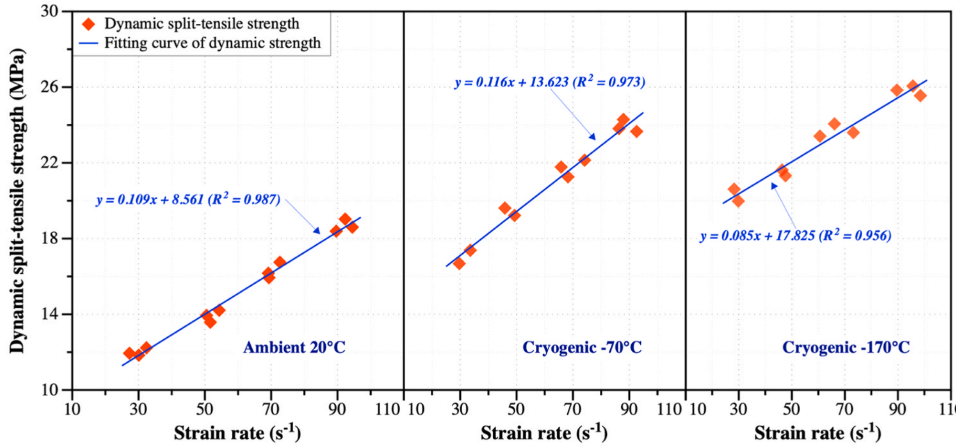

Fig. 13. Dynamic split-tensile strength of LHSC under different temperatures and strain rates.

27.27 s^{-1} to 94.48 s^{-1} . At −70 °C, the strength increased by 45.6 % from 16.7 MPa to 24.3 MPa within a similar strain rate range. At −170 °C, represented by the equation ($y = 0.085x + 17.825$, $R^2 = 0.956$), the strength enhancement was 30.4 % as the strain rate raised from 28.23 s^{-1} to 98.48 s^{-1} . Additionally, lowering the temperature from ambient to cryogenic conditions substantially enhanced the split-tensile strength, indicating the synergistic impacts of dynamic and cryogenic conditions on the tensile capacity of LHSC. As an illustration, when the strain rate was about 70 s^{-1}, the dynamic strength enhanced from 16.3 MPa (an average of multi strength values near 70 s^{-1}) at 20 °C to 21.7 MPa and 23.8 MPa as the temperatures decreased to −70 °C and −170 °C, respectively. These increments represent increases of 95.4 %, 143.0 %, and 158.4 % as compared to their static split-tensile strength under the same temperature conditions (as shown in Fig. 6(b)). It is noteworthy that while the slope of the strength-strain rate relationship increased from 20 °C to −70 °C, it slightly attenuated at −170 °C. This observation suggests that while ice formation within the LHSC matrix at ultra-low temperatures remarkably increased stiffness, the associated increase in brittleness and propensity for fracture under tensile loads at these temperatures may mitigate some of the ice-induced strengthening effects as the strain rate increased.

3.3.2. Dynamic failure mode and specimen impact process

Fig. 14 displays the characteristic dynamic split-tensile failure

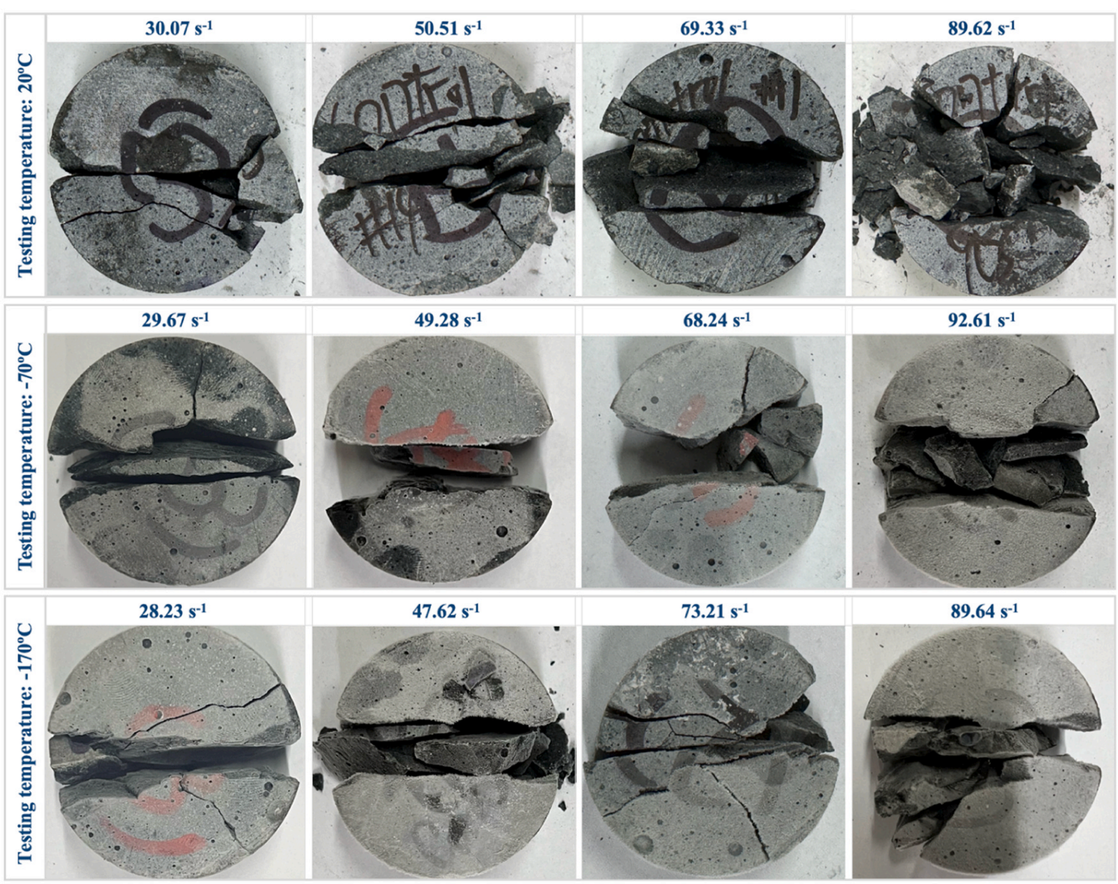

Fig. 14. Split-tensile failure patterns in LHSC specimens subjected to SHPB test at variable temperatures.

patterns in LHSC under various testing conditions. It can be clearly observed that the fracture predominantly occurred along the diametral direction, with the fracture surfaces oriented perpendicular to the end of the cylindrical specimens across all temperatures and strain rates. Notably, an increase in strain rate at a constant temperature typically resulted in more catastrophic fractures. As an illustration, at 20 °C and a strain rate of 30.07 s^{-1}, the specimen displayed two central diametral cracks with small triangular fracture zones at both ends. In contrast, at a higher rate of 89.62 s^{-1}, the cylinder scattered into more fragmented debris with enlarged triangular fracture areas. Besides, lowering temperatures from ambient to cryogenic conditions signally exacerbated the brittleness of LHSC, as evidenced by more pronounced cracking and complete disintegration of the specimens upon failure. As an illustration, at approximately 50 s^{-1}, the specimens at −70 °C and −170 °C showed clear diametral splits while retaining some coherent blocks, whereas the specimen at 20 °C exhibited extensive cracking with fewer residual bridging fragments. Furthermore, as the strain rate escalated, the emergence of wedge-shaped sections on both sides of the cylinders at −70 °C and −170 °C became more evident. The central tensile region also expanded and subsequently disintegrated, resulting in more severe triangular fragmentation.

Fig. 15 captures the dynamic split-tensile failure process of LHSC specimens at approximately 50 s^{-1} across three temperatures using a high-speed camera. At 0 μs, just before dynamic loading, no visible cracks were evident. By around 40 μs, the specimen at ambient 20 °C began to show an initial microcrack centrally aligned with the loading direction. In contrast, the specimens at −70 °C and −170 °C exhibited broader microcracks extending to the specimen edges, accompanied by several branching cracks and small triangular fracture zones. By 106 μs, the microcrack at 20 °C had widened into a primary macrocrack with emerging branch cracks, while at −70 °C and −170 °C, multiple distinct macrocracks were already evident, propagating more rapidly across the diameter. At 220 μs, the fragmentation was most severe at −170 °C, with the specimen nearly completely disintegrated into multiple pieces. The −70 °C specimen showed substantial cracking but retained some bridging segments, while the 20 °C specimen, although significantly cracked along the central plane, remained less fragmented. These sequences revealed that the brittleness of LHSC notably heightened under cryogenic conditions, leading to more rapid and pronounced crack propagation and coalescence, ultimately resulting in more severe fracture as compared to ambient conditions.

3.3.3. Strain rate sensitivity

Fig. 16 illustrates the aggregate data points of DIF_{ft} for LHSC specimens subjected to various strain rates and temperatures (20 °C, −70 °C, and −170 °C). The graph highlighted the significant rate sensitivity of the LHSC. At 20 °C, the DIF_{ft} increased from 1.432 at approximately 27.27 s^{-1} to 2.282 at 92.31 s^{-1}, demonstrating a nearly 60 % increase. As temperatures decreased to −70 °C, this sensitivity escalated, with DIF_{ft} rising from 1.867 at around 29.67 s^{-1} to 2.720 at 87.92 s^{-1}, marking an increase of over 45%. At the even extreme temperature of −170 °C, the DIF_{ft} reached its highest value, climbing from 2.169 to 2.829, a 30.4% increase within a comparable strain rate range. Moreover, at a given strain rate, for instance, at approximately 50 s^{-1}, the DIF_{ft} escalated progressively with decreasing temperature: from 1.668 (an average of multi DIF_{ft} values near 50 s^{-1}) at 20 °C to 2.175 at −70 °C, and further to 2.332 at −170 °C. These increases of 30.4% and 39.8%, respectively, underlined a growing tendency of DIF_{ft} as temperatures reduced, revealing a heightened strain rate sensitivity at cryogenic temperatures in comparison with DIF_{fc} of LHSC (as detailed in

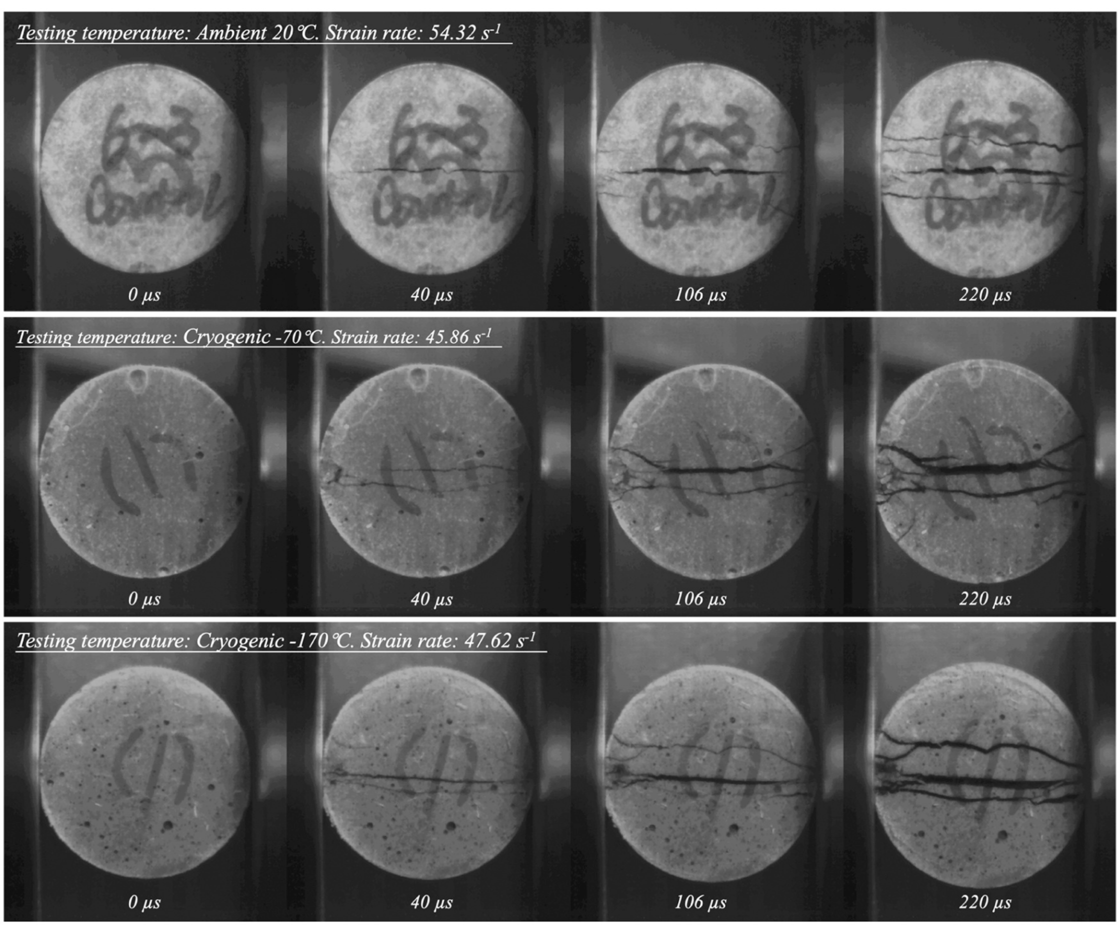

Fig. 15. Dynamic split-tensile failure process of LHSC specimens at different temperatures.

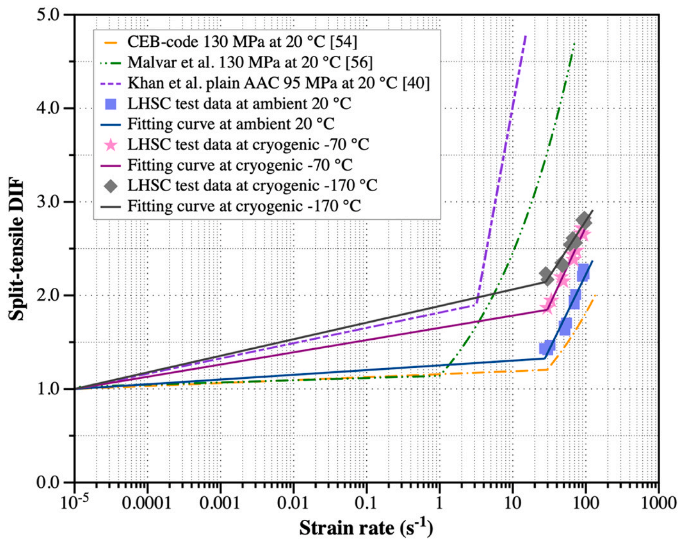

Fig. 16. Relationship between DIF_{ft} and strain rate of LHSC across various temperatures.

Section 3.2.4). This can be attributed to the differing fracture dynamics in tension versus compression. Cryogenic conditions, by promoting ice formation within the pore network, further stiffened the inherent brittleness of the geopolymer matrix, diminishing its capability to redistribute localized stresses. In split-tension scenarios, the material's failure relied on the rapid coalescence of cracks, where the ice-induced brittleness facilitated quicker crack propagation in comparison with compression scenarios. Here, the confinement effects and frictional forces can mitigate immediate damage, as shown in Figs. 10 and 14 comparing similar time intervals. This trend can also be observed from the research by Lian et al. [23] and Feng et al. [61,62].

The fitted equations for DIF_{ft} across different temperatures are specified in Eqs. (8)-(10), with the transition strain rates (approximately 30.07 s^{-1} at 20 °C, 29.67 s^{-1} at −70 °C, and 29.84 s^{-1} at −170 °C) closely matching the transition of 30 s^{-1} in CEB-FIP code. Fig. 16 also compares these findings with strain rate-DIF_{ft} relationships proposed by the CEB-FIP model [54], Malvar et al. [56], and Khan et al. [40] at 20 °C. While the DIF_{ft} curves at ambient aligned reasonably with CEB-FIP model, the latter two models appeared to overestimate the dynamic splitting behaviour of the LHSC, likely due to differences in binder composition, moisture content, curing processes, and testing environments specific to LHSC and other AACs.

$$DIF_{ft}\text{ at ambient}20\quad ^{\circ}C = \begin{cases} 0.051\log\dot{\varepsilon} + 1.253 & \dot{\varepsilon} \le 30.07s^{-1} \\ 1.601\log\dot{\varepsilon}\text{-}0.977 & \dot{\varepsilon} > 30.07s^{-1} \end{cases} R^2 = 0.936 \tag{8}$$

$$DIF_{ft}\text{ at cryogenic} - 70\quad ^{\circ}C = \begin{cases} 0.131\log\dot{\varepsilon} + 1.654\dot{\varepsilon} \le 29.67s^{-1} \\ 1.682\log\dot{\varepsilon}\text{-}0.632\dot{\varepsilon} > 29.67s^{-1} \end{cases} R^2 = 0.973 \tag{9}$$

$$DIF_{ft}\text{ at cryogenic} - 170\quad ^{\circ}C = \begin{cases} 0.177\log\dot{\varepsilon} + 1.887 & \dot{\varepsilon} \le 29.84s^{-1} \\ 1.191\log\dot{\varepsilon} + 0.415 & \dot{\varepsilon} > 29.84s^{-1} \end{cases} R^2 = 0.934 \tag{10}$$

In general, the effect of low temperatures on static and dynamic mechanical properties of LHSC is not identical. Under static conditions, the formation of ice within the pore structure improved matrix compactness, enhanced cohesive bonding, and hence elevated the compressive and split-tensile strengths. In contrast, under dynamic loading, while ice initially contributed to stiffness, the rapid stress wave propagation accelerated microcrack initiation and coalescence, leading to a more brittle response and reduced energy absorption. This difference arises from the combined effects between rate hardening and thermal induced microcracking, which dominates the fracture behaviour under high strain rates at cryogenic temperatures.

3.4. *Computed tomography (CT) scanning analysis*

X-ray CT stands out as a powerful, non-destructive method that produces cross-sectional views by measuring X-ray attenuation through materials and reconstructing this data into 3D images. This technique becomes increasingly prevalent in concrete research due to its ability to visualize the internal features like voids, cracks, and heterogeneities without damaging the specimens [63]. Unlike conventional microscopic methods, CT scanning can capture volumetric information in a single test, offering more holistic insights of the pore distribution and crack evolution under varying environmental conditions [64].

In this study, the Siemens Artis Pheno system, operating at 125 kV, was utilised to conduct CT scans on LHSC specimens at temperatures of 20 °C, −70 °C, and −170 °C. As illustrated in Fig. 17, the cross-sectional images provided a clear depiction of the changes in the matrix microstructure as the temperature decreased. At 20 °C, the matrix appeared dense with only a few large pores (light grey contours), correlating with superior mechanical properties of approximately 130 MPa in compressive strength and around 8.5 MPa in split-tensile strength, as previously detailed in Fig. 6. As temperatures decreased to −70 °C, the pores became visibly smaller, and scattered ice crystals (darker grey regions) started to form within the matrix. Concurrently, the emergence of microcracks (black and band-like lines) suggested a reduction in material's ability to withstand internal strains. Further reduction in temperature to −170 °C led to even smaller pores, while the ice crystals proliferated, intertwining extensively with the remaining pores. Microcracks became more numerous and extensive, creating nearly connected networks between the ice crystals and pores. Importantly, recent studies confirm that ice crystals filling pores increased contact areas between solid constituents and densify the matrix, which elevated the stiffness and compressive capacity [65]. In addition, the freezing of pore water generated interfacial crystallisation pressure that bound particles and microcracks together, leading to a temporary increase in strength [66]. However, this process simultaneously induced local tensile stresses along pore walls, which can promote the initiation of microcracks. These microscale transformations provide the mechanistic insights into the observed mechanical behaviour: while pore-filling ice enhances static stiffness and load-bearing capacity, the concurrent crystallisation pressure and microcracking accelerate embrittlement, particularly under dynamic loading where micro-defects propagate rapidly under stress [52,53]. The complex interaction among expanding ice networks, shrinking pore sizes, and escalating microcrack formation explains why LHSC shows increased strength and modulus, while its dynamic energy absorption and toughness deteriorate under cryogenic conditions.

4. Conclusions

The exploration of the extraterrestrial environments and the development of lunar bases present signal challenges for construction materials, necessitating enhanced performance under extreme conditions. The current study investigated the impacts of the multi-physics coupling, specifically the extreme temperatures and variable strain rates, on the mechanical properties of an innovative cement-free alkali-activated LHSC. The main conclusions are outlined as follows:

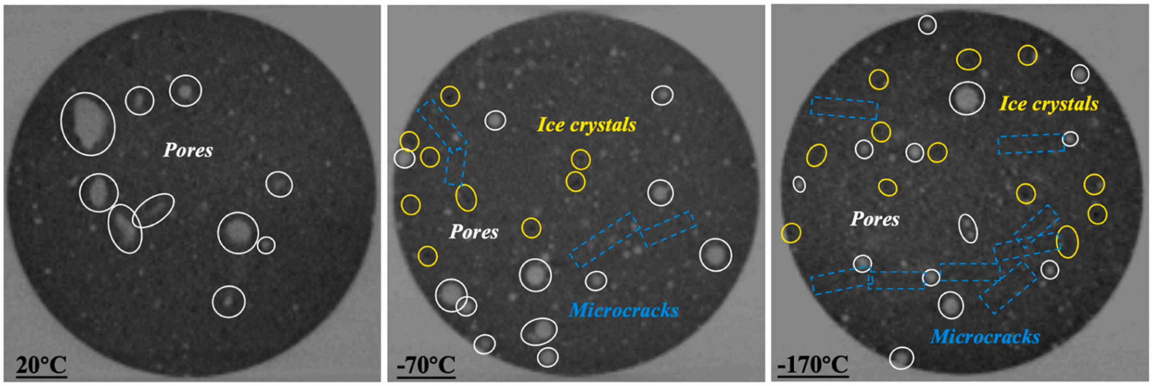

Fig. 17. Diametrical cross-sectional CT scan images of LHSC specimens at different temperatures.

1) LHSC exhibited notable increases in static strength as temperatures decreased from 20 °C to −170 °C. Specifically, the highest compressive strength was 153.6 MPa, and the split-tensile strength reached 9.2 MPa, indicating increases of 18.6% and 10.8%, respectively, relative to ambient conditions.
2) Under coupled temperature and strain rate conditions, the dynamic compressive strength of the LHSC significantly increased, surpassing static levels by up to 70.8% at −170 °C. Moreover, the energy absorption capacity enhanced with strain rate, showing an increase with rates but a decrease under cryogenic conditions. The dynamic MOE demonstrated enhanced sensitivity to rate changes at low temperatures, with peak performance around −70 °C before weakening at −170 °C.
3) At strain rates ranging from 30 s^{-1} to 90 s^{-1}, the dynamic split-tensile strength of the LHSC exhibited a consistent linear increase. The strength-rate sensitivity intensified from 20 °C to −70 °C whereas slightly diminished at −170 °C, suggesting that ice-induced stiffening improved tensile strength, but increased brittleness limited further enhancements.
4) Cryogenic conditions signally increased brittleness in both compressive and split-tensile SHPB tests, resulting in accelerated crack propagation and severe fragmentation. Higher strain rates led to finer debris, while triangular fracture zones became more pronounced at cryogenic temperatures.
5) The DIFs for both compression and split-tension demonstrated notable increases under high strain rates and decreasing temperatures, with cryogenic conditions signally raising rate sensitivity. The split-tensile DIF exceeded that of compression, revealing increased brittleness and faster crack propagation under tensile stress in cryogenic environments.
6) Microstructural analysis using CT scanning indicated that lowering temperatures from 20 °C to −170 °C progressively reduced pore sizes while promoting ice crystal formation, which interlaced with matrix voids and intensified microcracking. These transformations strengthened the matrix and improved the mechanical properties of LHSC, but they also heightened brittleness and reduced material deformability.

This study represents a critical and innovative exploratory effort toward understanding both the static and dynamic behaviour of the LHSC, leveraging substantial in-situ lunar regolith resources. Key insights have been gleaned concerning its strength development, damage mechanism, strain rate sensitivity, and microstructural evolutions under cryogenic temperatures. Nevertheless, the formulation employed did not use 100% lunar-derived materials, nor did the static and dynamic testing replicate the vacuum conditions prevalent on the lunar surface. Additionally, temperature fluctuations on the lunar surface, which occur repeatedly within a lunar day, introduce additional thermal cycling effects that can substantially affect long-term durability of lunar concrete. Future investigations will hence aim to overcome these limitations by integrating a 100% lunar material composition, simulating vacuum conditions, and incorporating temperature cycling. Moreover, future work will also focus on developing and calibrating advanced constitutive models, based on integrated experimental data and numerical simulations, to provide a unified, comprehensive prediction framework for strain rate and DIF behaviour of LHSC under diverse space conditions. Such efforts are expected to foster a more systematic understanding of the LHSC for prolonged extraterrestrial application.

CRediT authorship contribution statement

Jun Li: Writing – review & editing, Methodology, Conceptualization. **Ruizhe Shao:** Writing – original draft, Methodology, Investigation, Conceptualization. **Chengqing Wu:** Writing – review & editing, Visualization, Supervision, Resources, Methodology, Conceptualization. Kaiyi Chi: Investigation, Methodology. **Zizheng Yu**: Investigation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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